

WORKED EXAMPLE

Start from the saved forward pass

Keep the original parameters from Lessons 16-17 fixed throughout backpropagation.

$$x = (1, 2), \quad t = 1, \quad y = 0.5466967333$$

$$y_1^{(2)} = 0.5627638384, \quad y_2^{(2)} = 0.4537407133$$

$$w_{1,1}^{(3)} = 0.3, \quad w_{2,1}^{(3)} = -0.4, \quad b_1^{(3)} = 0.2$$

$$z_1^{(3)} = w_{1,1}^{(3)} y_1^{(2)} + w_{2,1}^{(3)} y_2^{(2)} + b_1^{(3)}$$

Our goal is all three output-parameter gradients: how sensitive this one example loss is to each parameter when the others stay fixed.

The handout calls its per-example loss L . We write ℓ to keep it distinct from a later batch mean J .

WORKED EXAMPLE

Delta measures sensitivity to preactivation

The backward calculation follows the same dependencies as the forward calculation.

$$w \longrightarrow z \longrightarrow y = \sigma(z) \longrightarrow \ell$$

$$\delta_1^{(3)} = \frac{\partial \ell}{\partial z_1^{(3)}}$$

$$\frac{\partial \ell}{\partial w} = \frac{\partial \ell}{\partial z} \frac{\partial z}{\partial w} = \delta \frac{\partial z}{\partial w}$$

Delta is a derivative with respect to z . A weight gradient is a derivative with respect to a weight. They are connected by the chain rule but are not usually the same number.

First calculate delta at the output; then multiply by each source activation to get that connection gradient. The bias has constant source activation 1.

WORKED EXAMPLE

Differentiate binary cross-entropy

Handout page 2, section 4.2; assume $0 < y < 1$.

$$\ell = -[t \ln y + (1 - t) \ln(1 - y)]$$

$$\frac{\partial \ell}{\partial y} = -\frac{t}{y} + \frac{1-t}{1-y}$$

$$= \frac{-t(1-y) + (1-t)y}{y(1-y)}$$

$$= \frac{-t + ty + y - ty}{y(1-y)} = \frac{y-t}{y(1-y)}$$

The plus sign in the second term comes from two negatives: the loss has a minus sign and the derivative of $(1-y)$ is -1 .

WORKED EXAMPLE

Cancel the sigmoid derivative once

Handout page 2, section 5.1: this shortcut needs sigmoid output and BCE loss.

$$\frac{\partial y}{\partial z} = y(1 - y)$$

$$\delta = \frac{\partial \ell}{\partial y} \frac{\partial y}{\partial z}$$

$$\delta = \frac{y - t}{y(1 - y)} y(1 - y) = y - t$$

Do not multiply by $y(1-y)$ again after using $\delta = y-t$. That sigmoid factor has already cancelled.

This is a per-example derivative. There is no division by the number of examples here. We will average parameter gradients once in the batch lesson.

WORKED EXAMPLE

Check both factors numerically

Target $t = 1$ means the BCE derivative with respect to y is $-1/y$.

$$\frac{\partial \ell}{\partial y} = -\frac{1}{0.5466967333} \approx -1.8291676885$$

$$\frac{\partial y}{\partial z} = 0.5466967333(1 - 0.5466967333)$$

$$\frac{\partial y}{\partial z} \approx 0.2478194151$$

$$\delta \approx (-1.8291676885)(0.2478194151) = -0.4533032667$$

$$\text{shortcut: } y - t = 0.5466967333 - 1 = -0.4533032667$$

The two routes agree. Negative delta means a tiny increase in this output preactivation decreases this example loss locally.

WORKED EXAMPLE

First output weight: use its source activation

Handout page 3, section 5.2. The source is hidden-layer-2 neuron 1.

$$\frac{\partial z_1^{(3)}}{\partial w_{1,1}^{(3)}} = y_1^{(2)}$$

$$\frac{\partial \ell}{\partial w_{1,1}^{(3)}} = \delta_1^{(3)} y_1^{(2)}$$

$$= (-0.4533032667)(0.5627638384)$$

$$\approx -0.2551026863$$

The source activation multiplies the weight in the forward weighted sum. That is why it becomes the local derivative with respect to the weight.

Use the activation 0.5627638384, not the hidden preactivation 0.2523866055.

WORKED EXAMPLE

Second output weight: keep its own source

This connection starts at hidden-layer-2 neuron 2.

$$\frac{\partial z_1^{(3)}}{\partial w_{2,1}^{(3)}} = y_2^{(2)}$$

$$\frac{\partial \ell}{\partial w_{2,1}^{(3)}} = \delta_1^{(3)} y_2^{(2)}$$

$$= (-0.4533032667)(0.4537407133)$$

$$\approx -0.2056821476$$

The current weight is -0.4, but this derivative uses its source activation, not the weight itself. The weight will matter when propagating a derivative back to the source neuron.

Keep the original -0.4 for the hidden-layer calculation in Lesson19. Do not update it yet.

WORKED EXAMPLE

Output bias: the source is constant one

Adding one unit to the bias adds one unit to the preactivation.

$$\frac{\partial z_1^{(3)}}{\partial b_1^{(3)}} = 1$$

$$\frac{\partial \ell}{\partial b_1^{(3)}} = \delta_1^{(3)}(1) = -0.4533032667$$

output parameter	one-example gradient
w1,1 in layer3	-0.2551026863
w2,1 in layer3	-0.2056821476
bias in layer3	-0.4533032667

The bias gradient equals delta because its multiplier is 1. This equality does not hold for the other two weight gradients.

WORKED EXAMPLE

What does a negative gradient predict?

Diagnostic experiment: vary only the output bias; this is not a training update.

$$\Delta b = +0.001, \quad \Delta \ell \approx \frac{\partial \ell}{\partial b} \Delta b$$

$$\Delta \ell \approx (-0.4533032667)(0.001) = -0.0004533033$$

actual fresh loss change ≈ -0.0004531794

The local estimate is close, not exact, for a finite perturbation. All other weights and biases are held fixed.

Return the bias to its original 0.2. Backpropagation in Lesson19 continues from the original parameter state and forward cache.

Central finite-difference checks of all three output gradients agree with the analytic results.

INDEPENDENT PRACTICE

Your turn: flip only the known target

Use the same original input (1,2), parameters and forward prediction; set $t = 0$.

1. Explain which forward quantities change when only the target changes.
2. Calculate BCE, the output delta and all three output-parameter gradients.
3. Explain the sign of the bias gradient using a tiny positive bias change with every other parameter fixed.

Do not update any parameters. This is the target-flip retry from numerical-practice N3.6.

WORKED ANSWER

Practice answer: prediction stays the same

Targets enter the loss; this feedforward model does not use them as inputs.

$$y = 0.5466967333, \quad \ell = -\ln(1 - y) \approx 0.7911939146$$

$$\delta = y - t = 0.5466967333 - 0 = 0.5466967333$$

$$\frac{\partial \ell}{\partial y} = \frac{1}{1 - y} \approx 2.2060286644$$

$$\delta \approx (2.2060286644)(0.2478194151) = 0.5466967333$$

All forward z and y values remain unchanged. The target changes the loss and the backward derivatives. Positive delta means increasing z locally increases the loss for target zero.

WORKED ANSWER

Practice answer: all three output gradients

Use the same two hidden activations and the new positive delta.

$$\frac{\partial \ell}{\partial w_{1,1}^{(3)}} = (0.5466967333)(0.5627638384) \approx 0.3076611521$$

$$\frac{\partial \ell}{\partial w_{2,1}^{(3)}} = (0.5466967333)(0.4537407133) \approx 0.2480585657$$

$$\frac{\partial \ell}{\partial b_1^{(3)}} = (0.5466967333)(1) = 0.5466967333$$

A tiny positive bias change now increases the loss: it raises the probability of class 1 when the known target is 0.

Checklist: distinguish delta from a weight gradient, use source activations, cancel sigmoid only once, and keep every parameter unchanged until all required gradients are ready.